Pathobiological determinants of atherosclerosis in youth risk scores are associated with early and advanced atherosclerosis.

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OBJECTIVES: Atherosclerosis begins in childhood and progresses during adolescence and young adulthood. The Pathobiological Determinants of Atherosclerosis in Youth Study previously reported risk scores to estimate the probability of advanced atherosclerotic lesions in young individuals aged 15 to 34 years using the coronary heart disease risk factors (gender, age, serum lipoprotein concentrations, smoking, hypertension, obesity, and hyperglycemia). In this study we investigated the relation of these risk scores to the early atherosclerotic lesions. METHODS: We measured atherosclerotic lesions in the left anterior descending coronary artery, right coronary artery, and abdominal aorta and the coronary heart disease risk factors in persons 15 to 34 years of age who died as a result of external causes and were autopsied in forensic laboratories. RESULTS: Risk scores computed from the modifiable risk factors were associated with prevalence of microscopically demonstrable lesions of atherosclerosis (American Heart Association grade 1) in the left anterior descending coronary artery and with the extent of the earliest detectable gross lesion (fatty streaks) in the right coronary artery and abdominal aorta. Risk scores computed from the modifiable risk factors also were associated with prevalence of lesions of higher degrees of microscopic severity (intermediate as well as advanced) in the left anterior descending coronary artery and with extent of lesions of higher degrees of severity (intermediate and raised lesions) in the right coronary artery and abdominal aorta. CONCLUSIONS: Risk scores calculated from traditional coronary heart disease risk factors to identify individual young persons with high probability of having advanced atherosclerotic lesions also are associated with earlier atherosclerotic lesions, including the earliest anatomically demonstrable atherosclerotic lesion. These results support lifestyle modification in youth to prevent development of the initial lesions and the subsequent progression to advanced lesions and, thereafter, to prevent or delay coronary heart disease.